



Mathematics
and Statistics

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Mathematics 2C03

Introduction to Differential Equations

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Lecture 13
Practice Test 1 solutions
Tuesday 6 October 2026

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Practice Test 1 solutions

1 Practice Test 1 solutions

- Q1: Academic-integrity declaration
- Q2: Classification
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Question 1: Academic-integrity declaration

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Question 2: Classifying an ODE

Classify the following differential equation according to type, order, and linearity, in that order.

$$-9x \left(\frac{dy}{dx} \right)^2 + \sin(63x)y = e^{7x}.$$

- A) ordinary, second order, linear
- B) ordinary, first order, linear
- C) partial, first order, linear
- D) partial, second order, non-linear
- E) ordinary, first order, non-linear
- F) ordinary, second order, non-linear
- G) partial, second order, linear
- H) partial, first order, non-linear

Solution to Question 2: Classifying an ODE

$$-9x \left(\frac{dy}{dx} \right)^2 + \sin(63x)y = e^{7x}.$$

- *Ordinary*: only the ordinary derivative $\frac{dy}{dx}$ appears.
- *First order*: the highest derivative is $\frac{dy}{dx}$, even though it is squared.
- *Nonlinear*: the term $\left(\frac{dy}{dx} \right)^2$ is nonlinear in a derivative of y . A coefficient such as $\sin(63x)$ multiplying y does not make an equation nonlinear.

Answer E: ordinary, first order, non-linear.

Question 3: Where does the theorem apply?

For which of the following points does one of the theorems given in class guarantee that the differential equation

$$y' = e^{x^2} \sqrt{100 - y^2}$$

possesses a unique solution through the given point?

- (i) $(-4, 14)$ (ii) $(-2, 10)$ (iii) $(3, 6)$

A) (ii) and (iii) only

B) none of them

C) all of them

D) (ii) only

E) (i) only

F) (i) and (iii) only

G) (iii) only

H) (i) and (ii) only

Solution to Question 3: Where does the theorem apply?

$$y' = f(x, y) = e^{x^2} \sqrt{100 - y^2}, \quad \frac{\partial f}{\partial y} = -\frac{e^{x^2} y}{\sqrt{100 - y^2}}.$$

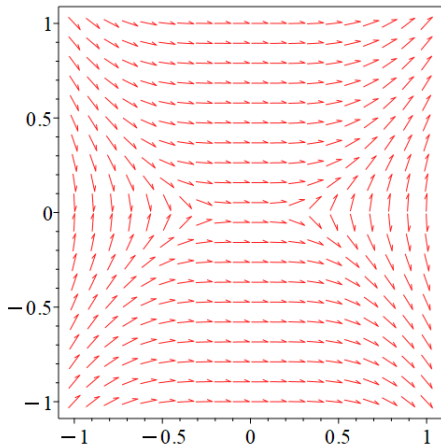
The **existence–uniqueness theorem** requires f and $\frac{\partial f}{\partial y}$ to be continuous on a rectangle containing the initial point in its interior.

- (i) $(-4, 14)$: $f(-4, 14)$ is not real. The theorem does not apply.
- (ii) $(-2, 10)$: every such rectangle includes values with $y > 10$, where the square root is not real. The theorem does not apply.
- (iii) $(3, 6)$: choose a small rectangle with $-10 < y < 10$ throughout. Both functions are continuous there, so the theorem guarantees a unique local solution.

Answer G: (iii) only. Note: Failure of the theorem's hypotheses does not by itself prove nonuniqueness.

Question 4: Identifying a direction field

Which differential equation corresponds to the following slope field?



A) $\frac{dy}{dx} = y^2$

B) $\frac{dy}{dx} = x^3$

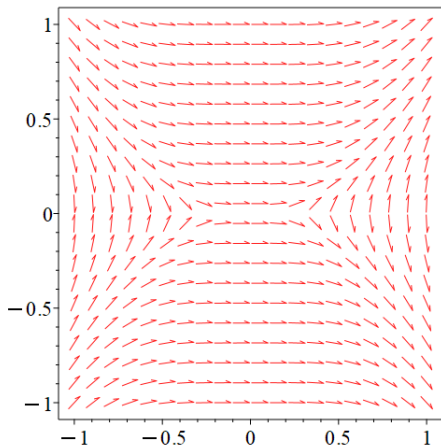
C) $\frac{dy}{dx} = x^3/y$

D) $\frac{dy}{dx} = x^2$

E) $\frac{dy}{dx} = y^3$

F) $\frac{dy}{dx} = y^3/x$

Solution to Question 4: Identifying a direction field



- Positive slopes in quadrants I and III, negative in II and IV.
- Neither y^2 nor x^2 can be negative.
- Neither x^3 nor y^3 changes sign across both axes.
- This leaves x^3/y or y^3/x .
- On the y -axis, away from the origin, the slopes are zero:

Answer C: $y' = x^3/y$.

Question 5: Long-term behaviour

Let $y(x)$ denote the unique solution of the initial-value problem

$$y' = -y^{14}(y^2 - 8y + 16), \quad y(0) = 2.$$

Using the phase line of this autonomous differential equation, find the value of $\lim_{x \rightarrow \infty} y(x)$. (DO NOT SOLVE)

- A) 4
- B) ∞
- C) none of these
- D) 0
- E) -4

- $y' = -y^{14}(y - 4)^2$.
- The equilibria are $y = 0$ and $y = 4$.
- Away from them, $y^{14} > 0$ and $(y - 4)^2 > 0$, so $y' < 0$.
- Starting at $y(0) = 2$, the solution moves downwards in $0 < y < 4$.
- By **uniqueness**, it cannot cross $y = 0$. Its limit is therefore 0.

Answer D: $\lim_{x \rightarrow \infty} y(x) = 0$.



Question 6: Solving a separable IVP

Suppose that $y(x)$ is the solution to the initial value problem

$$y' = y(3 - 20x^3), \quad y(1) = e^{85}.$$

Find $y(2)$.

A) e^{13}

B) e^{21}

C) e^9

D) e^{11}

E) e^{19}

F) e^{15}

G) e^{17}

H) e^7

For $y \neq 0$, **separate variables** and integrate:

$$\frac{1}{y} \frac{dy}{dx} = 3 - 20x^3,$$

$$\ln |y| = 3x - 5x^4 + C.$$

The initial condition gives

$$85 = 3 - 5 + C \quad \implies \quad C = 87.$$

Since the solution is positive at $x = 1$, take the positive branch, *i.e.*, replace $|y(x)|$ with $+y(x)$, so

$$y(x) = e^{3x - 5x^4 + 87},$$

$$y(2) = e^{6 - 80 + 87} = e^{13}$$

Answer A.

Question 7: Population growth with immigration

A deer population in a protected reserve grows naturally at a rate proportional to its current size, with proportionality constant $1/2$ per year. In addition, deer immigrate into the reserve at a constant rate of 20 deer per year. Initially, there are 100 deer. Let $P(t)$ denote the population t years later.

(a) According to this model, what is the initial rate of change $P'(0)$?

- A)** 20 deer per year
- B)** 50 deer per year
- C)** 70 deer per year
- D)** 100 deer per year
- E)** 120 deer per year

(b) What is $P(2 \ln 2)$?

- A)** 100 deer
- B)** 140 deer
- C)** 220 deer
- D)** 240 deer
- E)** 280 deer

Solution to Question 7(a): Population growth with immigration

For **part (a)**:

- Natural growth contributes $\frac{1}{2}P$ deer per year.
- Immigration contributes 20 deer per year.

Thus the IVP is

$$\frac{dP}{dt} = \frac{1}{2}P + 20, \quad P(0) = 100.$$

Now substitute the initial population into the ODE:

$$P'(0) = \frac{1}{2}(100) + 20 = 70$$

Answer to 7(a): C, 70 deer per year.

Solution to Question 7(b): Population growth with immigration

For **part (b)**:

- Convert to standard form: $P' - \frac{1}{2}P = 20$. Then **EMSID**:

E The **integrating factor** is $e^{\int (-1/2) dt} = e^{-t/2}$.

M, S Multiply by $e^{-t/2}$, then Simplify via the product rule:

$$\frac{d}{dt}(e^{-t/2}P) = 20e^{-t/2}.$$

I, D Integrate, then Divide by $e^{-t/2}$:

$$e^{-t/2}P = -40e^{-t/2} + C \quad \implies \quad P = -40 + Ce^{t/2}.$$

- Finally: $P(0) = 100$ gives $100 = -40 + C$, so $C = 140$.

$$P(2 \ln 2) = -40 + 140(2) = 240$$

Answer to 7(b): D, 240 deer.

Solution to Question 7(b): Population growth with immigration

Alternatively, for **part (b)**:

- Solve $P' = \frac{1}{2}P + 20$ by **separating variables**.

$$\frac{dP}{dt} = \frac{1}{2}(P + 40) \quad \Rightarrow \quad \frac{dP}{P + 40} = \frac{1}{2} dt.$$

- Integrate using $P(0) = 100$:

$$\int_{100}^{P(t)} \frac{dP}{P + 40} = \int_0^t \frac{1}{2} dt \quad \Rightarrow \quad \ln\left(\frac{P(t) + 40}{140}\right) = \frac{t}{2}.$$

- Here $P \geq 0$ is a population, so $P + 40 > 0$. Exponentiating gives

$$P(t) + 40 = 140e^{t/2} \quad \Rightarrow \quad P(t) = 140e^{t/2} - 40.$$

- Again, $P(2 \ln 2) = 240$ deer. *Answer to 7(b): D.*

Question 8: Solving a homogeneous nonlinear IVP

Use the method of homogeneous equations to solve the initial-value problem $(6x^2 + 2xy)y' = 9x^2 + 6xy + 2y^2$, $y(1) = -6$. Find $y(e^3)$.

A) $3e^3$

B) $-6e^3$

C) $-9e^3$

D) -9

E) $-9e$

- Divide by x^2 near $x = 1$:

$$\left(6 + 2\frac{y}{x}\right)y' = 9 + 6\frac{y}{x} + 2\left(\frac{y}{x}\right)^2.$$

- This is **homogeneous**: y' is a function of y/x alone. Set

$$u = y/x, \quad y = xu, \quad y' = u + xu'.$$

- Substitute and simplify:

$$\begin{aligned}(6 + 2u)(u + xu') &= 9 + 6u + 2u^2, \\ 2(u + 3)xu' &= 9.\end{aligned}\tag{*}$$

- There is no constant solution $u \equiv c$: it would give $0 = 9$.

Solution to Question 8: Solving a homogeneous nonlinear IVP

- The transformed equation (*) separates: $2(u+3)du = (9/x)dx$.
Integrate:

$$(u+3)^2 = 9\ln x + C.$$

- Since $u(1) = -6$, $C = (-6+3)^2 = 9$. Thus
$$(u+3)^2 = 9(1+\ln x).$$

- Initially $u+3 = -3 < 0$, so take the negative root:

$$u = -3 - 3\sqrt{1+\ln x}, \quad y = x(-3 - 3\sqrt{1+\ln x}).$$

- On this branch, $x > e^{-1}$. In particular,

$$y(e^3) = e^3(-3 - 3\sqrt{4}) = -9e^3$$

Answer C.

Question 9: Solving a Bernoulli IVP

Solve the following initial value problem.

$$tx' - (5 + t)x = t^{-4}x^2, \quad x(1) = \frac{1}{4}.$$

Find $x(1 + \ln 2)$.

A) $\frac{1}{4}(1 + \ln 2)^5$

B) $\frac{2}{3}(1 + \ln 2)^5$

C) $\frac{2}{7}(1 + \ln 2)^5$

D) $\frac{2}{9}(1 + \ln 2)^5$

E) $2(1 + \ln 2)^5$

- Divide by t to obtain

$$x' - \left(\frac{5}{t} + 1\right)x = t^{-5}x^2.$$

- This is **Bernoulli** with $r = 2$,
 $p(t) = -(5/t + 1)$, $f(t) = t^{-5}$.
- Since $x(1) \neq 0$, set $v = x^{1-r} = x^{-1}$. Then
 $v' = -x^{-2}x'$.
- Multiply the equation for x by $-x^{-2}$:

$$\underbrace{-x^{-2}x'}_{v'} + \left(\frac{5}{t} + 1\right) \underbrace{x^{-1}}_v = -t^{-5}.$$

Solution to Question 9: Solving a Bernoulli IVP

- We have a linear IVP:

$$v' + \left(\frac{5}{t} + 1\right)v = -t^{-5}, \quad v(1) = 4.$$

- Solve the linear IVP for v using EMSID:

E $\mu(t) = e^{\int (5/t+1) dt} = t^5 e^t$, since $t > 0$ near $t = 1$.

M, S $\frac{d}{dt}(t^5 e^t v) = -e^t$.

I, D $t^5 e^t v = -e^t + C$, hence $v = (-1 + Ce^{-t})/t^5$.

- Using $v(1) = 4$ gives $4 = -1 + C/e$, so $C = 5e$.

- Return to $x = 1/v$: $x(t) = \frac{t^5}{-1 + 5e^{1-t}}$.

- At $t = 1 + \ln 2$, $e^{1-t} = 1/2$, so

$$x(1 + \ln 2) = \frac{(1 + \ln 2)^5}{-1 + 5/2} = \frac{2}{3}(1 + \ln 2)^5$$

Answer (B)

Question 10: Solving an exact IVP

Solve the initial value problem

$$(4xy^2 + 4y + 6x^2y) dx + (4x^2y + 4x + 2x^3) dy = 0, \quad y(1) = 1.$$

Find $[y(2)]^2 + 3y(2)$.

- A) -4
 - B) -2
 - C) -5
 - D) -1
 - E) 2
 - F) 1
 - G) 0
 - H) -3
- Let $M = 4xy^2 + 4y + 6x^2y$ and $N = 4x^2y + 4x + 2x^3$.
 - The form is **exact**: $M_y = 8xy + 4 + 6x^2 = N_x$.
 - Seek a **potential** with $\Phi_x = M$ and $\Phi_y = N$.
 - Integrate $\Phi_x = M$ with respect to x :
$$\Phi(x, y) = 2x^2y^2 + 4xy + 2x^3y + h(y).$$

continued...

Solution to Question 10: Solving an exact IVP

- Differentiate with respect to y and compare with N :

$$\Phi_y = 4x^2y + 4x + 2x^3 + h'(y) = N.$$

- Thus $h'(y) = 0$. We can take $h = 0$, giving

$$\Phi(x, y) = 2x^2y^2 + 4xy + 2x^3y.$$

- The **potential** satisfies $\Phi(x, y) = C$. The initial condition gives

$$C = \Phi(1, 1) = 2 + 4 + 2 = 8.$$

- At $x = 2$,

$$8[y(2)]^2 + 24y(2) = 8.$$

- Dividing by 8 gives

$[y(2)]^2 + 3y(2) = 1$

Answer F.

Question 11: An integrating factor

Consider the initial-value problem

$$(6x^4 - \frac{1}{2}xy^2 + 2y) dx + (x^2y - x) dy = 0, \quad y(4) = 12.$$

Which of the following is an integrating factor depending only on x that makes the differential form exact near $(4, 12)$?

A) $\mu(x) = x$

B) $\mu(x) = x^{-3}$

C) $\mu(x) = e^{-3x}$

D) $\mu(x) = e^x$

E) $\mu(x) = e^{-x}$

F) $\mu(x) = x^3$

G) $\mu(x) = e^{3x}$

■ $M = 6x^4 - \frac{1}{2}xy^2 + 2y, \quad N = x^2y - x.$

■ $M_y = -xy + 2, \quad N_x = 2xy - 1, \text{ not exact.}$

■ $\frac{M_y - N_x}{N} = \frac{3 - 3xy}{x(xy - 1)} = -\frac{3}{x}, \text{ depends only on } x.$

■ Near $(x, y) = (4, 12), \quad N \neq 0 \text{ and } x > 0, \text{ so}$

$$\mu(x) = e^{\int (-3/x) dx} = x^{-3}$$

Answer B.

■ **Check:**

$$\frac{\partial}{\partial y}(\mu M) = \frac{\partial}{\partial y} \left(6x - \frac{y^2}{2x^2} + \frac{2y}{x^3} \right) = -\frac{y}{x^2} + \frac{2}{x^3}$$

$$\frac{\partial}{\partial x}(\mu N) = \frac{\partial}{\partial x} \left(\frac{y}{x} - \frac{1}{x^2} \right) = -\frac{y}{x^2} + \frac{2}{x^3}$$

Good luck!

Solving linear 1st order ODEs

Nonhomogeneous first order linear equation in standard form:

$$\frac{dy}{dx} + P(x)y = f(x) \quad (*)$$

EMSID *method to find the general solution:*

- 1 **E**valuate the integrating factor $e^{\int P(x) dx}$
- 2 **M**ultiply both sides of (*) by the integrating factor
- 3 **S**implify using product rule:

$$\frac{d}{dx} \left(e^{\int P(x) dx} y \right) = e^{\int P(x) dx} f(x) \quad (**)$$

- 4 **I**ntegrate both sides of (**) (but add only one constant).
- 5 **D**ivide both sides by the integrating factor (to isolate y).

Note: "EMSID" isn't standard terminology; it's just a suggestion to help you remember the method.